

## Module - I

### Set Theory, Relations, Functions & Natural Number

It is a branch of Computer Science that deals with mathematical part such as set, graph and sequence. Its applications are data structure, cryptography and many more.

#### Data

##### Discrete

(Whole Number Category)

Example:

- No. of students in class.
- No. of departments.

##### Continuous

(Decimal Number, Scale)

Example:

- Height of any person.
- Weight of any object.

## # Set Theory

It is a branch of Mathematics that study set, which are collection of distincts objects. Set theory is introduced by German Mathematician Georg Cantor in 1874.

Set is a collection of well defined objects. The object constitute set are called elements or member of set.

Example:  $N = \{1, 2, 3, 4, 5, \dots\}$

$V = \{a, e, i, o, u\}$

## Common Set Representations:

$\mathbb{N}$  - Set of all natural numbers.

$\mathbb{Z}$  - Set of all integer numbers.

$\mathbb{Q}$  - Set of all rational numbers.

$\mathbb{R}$  - Set of all real numbers.

$\mathbb{C}$  - Set of all complex numbers.

## Methods of Representing Sets:

### 1. Roster Method:

A set is described by listing elements separated by commas within curly braces.

Example:  $A = \{2, 4, 6, 8\}$ .

### 2. Set-Builder Method:

A set is described by characterizing properties of its elements. The set is written as  $\{x : P(x) \text{ holds}\}$ , read as the set of all  $x$  such that  $P(x)$  holds.

$$A = \{1, 4, 9, 16, 25\}$$

$$B = \{x : x = n^2, n \in \mathbb{N}\}$$

## Types of Sets

### i. Empty / Null / Void Set:

A set which has no elements is called an empty set.

$$A = \{x \in \mathbb{R} : x^2 = -2\} = \emptyset$$

$$B = \{x \in \mathbb{N} : 6 < x < 7\} = \emptyset.$$

### ii. Singleton Set:

A set containing a single element is known as a singleton set.

$$A = \{x : x \text{ is an even prime number}\} = \{2\}.$$

### iii. Finite Set:

A set is finite if its elements can be listed (counted).

$$A = \{x \in \mathbb{N} : x \leq 50\}.$$

### iv. Infinite Set:

A set whose elements cannot be listed or counted.

$$A = \{x \in \mathbb{N} : x = 2n\}.$$

### \* Cardinal Number:

The order of a finite set or the number of elements in a finite set. It is represented by  $n(A)$ .

$$A = \{2, 3, 4, 5\}$$

$$n(A) = 4.$$

### v. Equal Sets:

Two sets  $A$  &  $B$  are said to be equal if every element of  $A$  is in  $B$  and every element of  $B$  is a member of  $A$  — i.e.  $A = B$  &  $n(A) = n(B)$ .

$$A = \{1, 2, 3, 4\}, B = \{4, 2, 3, 1\}.$$

### vi. Equivalent Sets:

Two finite sets  $A$  &  $B$  are equivalent if their cardinal numbers are the same i.e.  $n(A) = n(B)$ .

$$A = \{1, 2, 3, 4\}, B = \{5, 6, 7, 8\}.$$

### vii. Subset & Superset:

If every element of set  $A$  is also an element of set  $B$ , then  $A$  is a subset of  $B$  (i.e.  $A \subseteq B$ ) and  $B$  is a superset of  $A$  (i.e.  $B \supseteq A$ ).

$$A = \{1, 3, 5\}, B = \{1, 2, 3, 4, 5, 6, 7\}.$$

viii. Proper Subset:

If  $A$  is a subset of  $B$  ( $A \subseteq B$ ) and  $A \neq B$ , then  $A$  is a proper subset of  $B$  ( $A \subset B$ ).

$$A = \{1, 3, 5\}, B = \{1, 2, 3, 5, 7, 9\}.$$

\* Equality of Set:

If  $A$  is a subset of  $B$  and  $B$  is a subset of  $A$ , then  $A = B$ .

ix. Universal Set:

A set that contains all sets in a given context is called a universal set ( $U$ ).

$$A = \{2, 4, 6\}$$

$$B = \{1, 3, 7\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

Note: Universal set ( $U$ ) is the largest superset and Null set ( $\emptyset$ ) is the smallest subset.

x. Power Set:

Let  $A$  be a finite set then the set of all subsets of  $A$  is called the power set  $P(A)$ . In this, total elements  $= 2^n$ .

$$A = \{a, b, c\} \Rightarrow |P(A)| = 2^3 = 8.$$

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}.$$

## # Theorems of Set Theory

### Theorem - I:

Every set is a subset of itself.

### Theorem - II:

The empty set ( $\emptyset$ ) is a subset of every set.

### Theorem - III:

The total number of subsets of a finite set containing  $n$  elements is  $2^n$ .

Q. Represent the following set by the use of set builder method?

$$A = \{2, 4, 6, 8\}$$

$$B = \{5, 7, 9, 13, 17, 19\}$$

Sol<sup>n</sup>:

$$A = \{x : x \text{ is an even number, } 2 \leq x \leq 8\}$$

$$\& B = \{x : x \text{ is odd number, } 5 \leq x \leq 19\}.$$

## # Set Operations

### 1. Union ( $\cup$ ):

$$A = \{1, 2, 3\}$$

$$B = \{4, 5, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

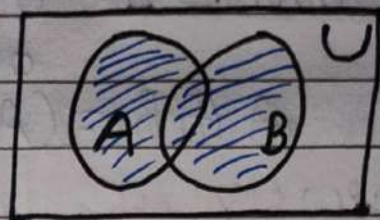


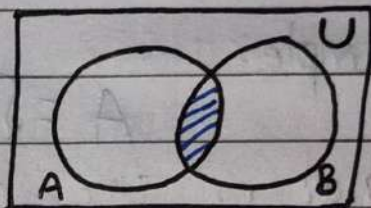
Fig.: Venn Diagram.

### 2. Intersection ( $\cap$ ):

$$A = \{1, 2, 3, 4\}$$

$$B = \{2, 4, 5\}$$

$$A \cap B = \{2, 4\}$$

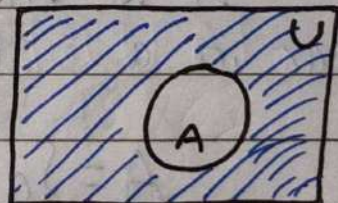


### 3. Compliment ( $'$ or $^c$ or $-$ ):

$$U = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A = \{2, 4, 6\}$$

$$A^c = \{1, 3, 5, 7\}$$



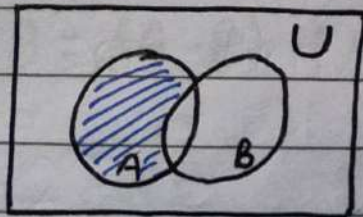
### 4. Difference ( $-$ ):

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$B = \{1, 2, 5, 6\}$$

$$A - B = \{3, 4\}$$

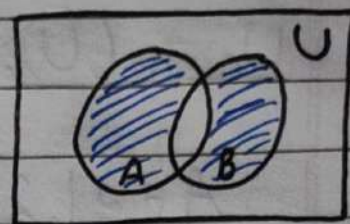
$$[B - A = \emptyset]$$



## 5. Symmetric Difference ( $\oplus$ or $\Delta$ ):

$$\rightarrow A \oplus B = (A \cup B) - (A \cap B)$$

$$\text{or}$$
$$A \oplus B = (A - B) \cup (B - A)$$



## # Multi-Set

It allows same type of multiple elements in the set.

Example:

$$A = \{1, 2, 2, 2, 3, 3, 4, 4\}$$

$$n(A) = 4.$$

Multiplicity:

It is the number of times an element appears in a set.

$$A = \{a, a, b, b, c, c\}$$

$$A = \{2.a, 2.b, 2.c\}$$

## Operations of Multi-set:

■  $A = \{a, a, a, b, b, c\}$  &  $B = \{a, a, b, d\}$

$$A = \{3.a, 2.b, 1.c\} \text{ \& } B = \{2.a, 1.b, 1.d\}$$

i. Union (U):

$$A \cup B = \{3.a, 2.b, 1.c, 1.d\}$$

ii. Intersection (∩):

$$A \cap B = \{2.a, 1.b\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

iii. Difference (-):

$$A - B = \{x : x \in A, x \notin B\}$$

$$B - A = \{x : x \in B, x \notin A\}$$

$$A - B = \{1.a, 1.b, 1.c\}$$

$$B - A = \{1.d\}$$

iv. Symmetric Difference  $[A \oplus B = (A - B) \cup (B - A)]$

$$A \oplus B = \{1.a, 1.b, 1.c, 1.d\}$$

v. Sum (+):

$$A + B = \{5.a, 3.b, 1.c, 1.d\}.$$

# # Principle of Inclusion & Exclusion

## Formulas:

For 2 sets,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For 3 sets,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

## Numericals:

Q. What is the cardinality of the set of integers such that:  $z = \{x \mid 1 \leq x \leq 123, x \text{ is divisible by 2 and 3}\}$ ?

Sol<sup>n</sup>:

$$|A \cup B| = \left\lfloor \frac{123}{2} \right\rfloor + \left\lfloor \frac{123}{3} \right\rfloor - \left\lfloor \frac{123}{6} \right\rfloor$$

$$= 61 + 41 - 20$$

$$= 82.$$

Q. What is the cardinality of the set of integers such that:  $z = \{x \mid 1 \leq x \leq 123, x \text{ is divisible by 2, 3 & 5}\}$ ?

Sol<sup>n</sup>:

$$|A \cup B \cup C| = \left| \frac{123}{2} \right| + \left| \frac{123}{3} \right| + \left| \frac{123}{5} \right|$$

$$- \left| \frac{123}{6} \right| - \left| \frac{123}{15} \right| - \left| \frac{123}{10} \right| + \left| \frac{123}{30} \right|$$

$$= 61 + 41 + 24 - 20 - 8 - 12 + 4$$

$$= 90.$$

Q. In a class of 400 students, 125 have C language, 85 take DS, 65 take COA, 50 take C & DS, 35 take DS & COA, 30 take C & COA, and 15 take all three. How many students take no course?

Sol<sup>n</sup>:

Given,  $|A| = 125$

$$|B| = 85$$

$$|C| = 65$$

$$|A \cap B| = 50$$

$$|B \cap C| = 35$$

$$|A \cap C| = 30$$

$$|A \cap B \cap C| = 15$$

Now,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

$$|A \cup B \cup C| = 125 + 85 + 65 - 50 - 35 - 30 + 15$$

$$= 175.$$

Hence, the number of student that haven't taken any course ~~is~~  $(400 - 175)$  i.e. 225.

Q. In a survey of 200 medical student, 50 use college library, 40 use own library, 30 use other sources, 20 use both college and own, 15 use own and other, 13 use college and other, and 10 use all three. How many use at least one source?

Sol<sup>n</sup>:

Given,  $|U| = 200$

$|A| = 50$

$|B| = 40$

$|C| = 30$

$|A \cap B| = 20$

$|B \cap C| = 15$

$|A \cap C| = 13$

$|A \cap B \cap C| = 10$

Now,

$$|A \cup B \cup C| = 50 + 40 + 30 - 20 - 15 - 13 + 10$$

$$= 82.$$

## # Ordered Pair

An ordered pair is a pair of elements enclosed in parentheses where the order of elements matters.

For ordered pair  $(a, b)$ ,  $a$  is the first element and  $b$  is the second element.

Note:  $(a, b) \neq (b, a)$ ,  $a \neq b$ .

## # Cartesian Product

The cartesian product of two sets  $A$  &  $B$  (denoted by  $A \times B$ ) is the set of all possible ordered pairs  $(a, b)$  such that  $a \in A$  and  $b \in B$ .

Example:

$$A = \{1, 2, 3\} \quad \& \quad B = \{1, 2\}$$

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

$$B \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

## # Properties of Cartesian Product

The properties of Cartesian Product for four sets  $A, B, C$  &  $D$ .

1.  $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$
2.  $(A - B) \times C = (A \times C) - (B \times C)$
3.  $(A \cup B) \times C = (A \times C) \cup (B \times C)$
4.  $A \times (B \cap C) = (A \times B) \cap (A \times C).$

Proof:

$$\underline{(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)}$$

Let  $(x, y) \in (A \cap B) \times (C \cap D)$ , that means  $x \in (A \cap B)$  and  $y \in (C \cap D)$ .

i.e.  $(x \in A \text{ and } x \in B)$

&  $(y \in C \text{ and } y \in D)$

So,  $(x, y) \in (A \text{ and } C)$  &  $(x, y) \in (B \text{ and } D)$

$\Rightarrow (x, y) \in (A \times C) \text{ and } (x, y) \in (B \times D)$

$\Rightarrow (x, y) \in (A \times C) \cap (B \times D).$